

Unit III

GROUP THEORY

OBJECTIVES:

After going through this unit, you will be able to know:

- Binary Operation
- Definition of Group, semi group, Monoid
- Permutation groups
- Cosets and Lagrange's theorem
- Homomorphism, Isomorphism and Automorphism of Groups
- Rings, integral domains and field.

INTRODUCTION:

In this chapter, we will study, binary operation as a function, and two more algebraic structures, semigroups and groups. They are called an algebraic structure because the operations on the set define a structure on the elements of that set. We also define the notion of a homomorphism and product and quotients of groups and semigroup.

BINARY OPERATION

A binary operation on a set A is an everywhere defined function $f: A \times A \rightarrow A$, generally the operation is denoted by $*$ on A , then $a * b \in A \quad \forall a, b \in A$.

Properties of binary operation : Let $*$

be a binary operation on a set A ,

Then $*$ satisfies the following

properties, namely

- Closure property
- Associative property
- Identity Property
- Inverse property
- Commutative property etc.

SEMIGROUP

A non-empty set S together with a binary operation $*$ is called as a semigroup if –

- i) binary operation $*$ is closed
 - ii) binary operation $*$ is associative
- we denote the semigroup by $(S, *)$

Commutative Semigroup :- A semigroup $(S, *)$ is said to be

commutative if $*$ is commutative i.e. $a * b = b * a \forall a \in S$

- Examples :**
- 1) $(\mathbb{Z}, +)$ is a commutative semigroup
 - 2) The set $P(S)$, where S is a set, together with operation of union is a commutative semigroup.
 - 3) $(\mathbb{Z}, -)$ is not a semigroup
The operation subtraction is not associative

IDENTITY ELEMENT :

An element e of a semigroup $(S, *)$ is called an identity element if $e * a = a * e = a \forall a \in S$

Monoid A non-empty set M together with a binary operation $*$ defined on it, is called as a monoid if –

- i) binary operation $*$ is closed
 - ii) binary operation $*$ is associative and
 - iii) $(M, *)$ has an identity.
- i.e. A semi group that has an identity is a monoid.

A non-empty set G together with a binary operation $*$ defined on it is called a group if

- (i) binary operation $*$ is close,
 - (ii) binary operation $*$ is associative,
 - (iii) $(G, *)$ has an identity,
 - (iv) every element in G has inverse in G ,
- We denote the group by $(G, *)$

Commutative (Abelian Group : A group $(G, *)$ is said to be commutative if $*$ is commutative. i.e. $a * b = b * a \forall a, b \in G$.

Cyclic Group : If every element of a group can be expressed as the power

The element is called as generator of the group.

If G is a group and a is its generator then we write $G = \langle a \rangle$

For example consider

$G = \{1, -1, i, -i\}$. G is a group under the binary operation of multiplication. Note that $G = \langle i \rangle$. Because

$$a = \{i, i^2, i^3, i^4\} = \{i, -1, -i, 1\}$$

SUBSEMI GROUP :

Let $(S, *)$ be a semigroup and let T be a subset of S . If T is closed under operation $*$, then $(T, *)$ is called a subsemigroup of $(S, *)$.

Submonoid : Let $(S, *)$ be a monoid with identity e , and let T be a non- empty subset of S . If T is closed under the operation $*$ and $e \in T$, then $(T, *)$ is called a submonoid of $(S, *)$.

Subgroup : Let $(G, *)$ be a group. A subset H of G is called as subgroup of G if $(H, *)$ itself is a group.

Necessary and Sufficient Condition for subgroup : Let $(G; *)$ be a group. A subset H of G is a subgroup of G if and only if $\forall a, b \in H \ a * b^{-1} \in H$

PERMUTATION GROUP

Definition : A permutation on n symbols is a bijective function of the set

$A = \{1, 2, \dots, n\}$ onto itself. The set of all permutations on n symbols is

denoted by S_n . If α is a permutation on n symbols, then α is completely

determined by its values $\alpha(1), \alpha(2), \dots, \alpha(n)$. We use following notation

to denote $\alpha \begin{pmatrix} 1 & 2 & 3 & \dots & n \\ \alpha(1) & \alpha(2) & \alpha(3) & \dots & \alpha(n) \end{pmatrix}$.

For example $\alpha \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 3 & 1 & 2 & 4 \end{pmatrix}$ denotes the permutation on the 5 symbols

$(1, 2, 3, 4, 5)$. α maps 1 to 5, 2 to 3, 3 to 1, 4 to 2 and 5 to 4.

Product of permutation : - Let $A = \{1, 2, 3, 4\}$

Let $\alpha \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 4 & 1 \end{pmatrix}$ and $\beta \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix}$.

Then $\alpha \circ \beta = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix}$

$$i_1, i_2, \dots, i_n \alpha(i_1) = i_2, \alpha(i_2) = i_3 \dots \alpha(i_n) = i_1.$$

Example : Consider following permutation

$$\begin{aligned} \text{i)} \quad & \alpha \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 3 & 4 & 1 & 6 & 5 \end{pmatrix}. \text{ It can be expressed as a product of cycles -} \\ & \alpha \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} \begin{pmatrix} 5 & 6 \\ 6 & 5 \end{pmatrix} = (1 \ 2 \ 3 \ 4)(5 \ 6) \end{aligned}$$

Transposition :

A cycle of length two is called transposition.

For example following permutation can be expressed as a product of transpositions.

$$\begin{aligned} & \alpha(1837)(25)(46) \\ \therefore & \alpha(18)(13)(17)(25)(46) \end{aligned}$$

Even (odd) Permutation -

Let $A = \{1, 2, \dots, n\}$. A permutation $\alpha \in S_n$ is even or odd according to whether it can be expressed as the product of an even number of transpositions or the product of an odd number of transpositions respectively.

For example we can consider following permutation :

$$\begin{aligned} \alpha &= (1 \ 4 \ 5)(2 \ 3) \\ \alpha &= (1 \ 4)(1 \ 5)(2 \ 3) \\ &= \text{odd no. of transpositions so } \alpha \text{ is odd permutation} \end{aligned}$$

Example 1 : Show that $*$ defined as $x * y = x$ is a binary operation on the set of positive integers. Show that $*$ is not commutative but is associative.

Solution : Consider two positive integers x and y . By definition $x * y = x$ which is a positive integer. Hence $*$ is a binary operation.
For commutativity : $x * y = x$ and $y * x = x$. Hence $x * y \neq y * x$ in general $\therefore *$ is not commutative.

But $x * (y * z) = x * y = x$ and $(x * y) * z = x * z = x$. Hence $x * (y * z) = (x * y) * z$. $\therefore *$ is associative

Example 2 : Let I be the set of integers and Z_m be the set of equivalence classes generated by the equivalence relation “congruent modulo m ” for

- Write the sets Z_3 and Z_6
- Show that the algebraic systems $(Z_m, +_m)$ and $(Z_m, \times_m)$ are monoids.
- Find the inverses of elements in Z_3 and Z_4 with respect to $+_3$ and $\times_4$ respectively.

Solution :

- Z_3 for $(Z_3, +_3) = \{[0], [1], [2]\}$
 Z_6 for $(Z_6, +_6) = \{[0], [1], [2], [3], [4], [5]\}$
 Z_3 for $(Z_3, \times_3) = \{[0], [1], [2]\}$
 Z_6 for $(Z_6, \times_6) = \{[0], [1], [2], [3], [4], [5]\}$

Example 3 : Determine whether the following set together with the binary operation is a semigroup, a monoid or neither. If it is a monoid, specify the identity. If it is a semigroup or a monoid determine whether it is commutative.

- $A =$ set of all positive integers.
 $a * b = \max\{a, b\}$ i.e. bigger of a and b
- Set $S = \{1, 2, 3, 6, 12\}$ where $a * b = G.C.D.(a, b)$
- Set $S = \{1, 2, 3, 6, 9, 18\}$ where $a * b = L.C.M.(a, b)$
- Z , the set of integers, where $a * b = a + b - ab$
- The set of even integers E , where $a * b = \frac{ab}{2}$
- Set of real numbers with $a * b = a + b + 2$
- The set of all $m \times n$ matrices under the operation of addition.

Solution :

- $A =$ set of all positive integers. $a * b = \max\{a, b\}$ i.e. bigger of a and b .

Closure Property: Since $\max\{a, b\}$ is either a or $b \therefore a * b \in A$. Hence closure property is verified.

Associative Property :

$$\begin{aligned} \text{Since } a * (b * c) &= \max\{a, b * c\} = \max\{a, \max\{b, c\}\} \\ &= \max\{a, \{b, c\}\} = (a * b) * c \end{aligned}$$

$\therefore *$ is associative.

$\therefore (A, *)$ is a semigroup.

Existence of identity : $1 \in A$ is the identity because

$$1 * a = \max\{1, a\} = a \quad \forall a \in A$$

$\therefore (A, *)$ is a monoid.

Commutative property : Since $\max\{a, b\} = \max\{b, a\}$ we have $a * b = b * a$ Hence $*$ is commutative.

ii) Set $S = \{1, 2, 3, 6, 12\}$ where $a * b = G.C.D. (a, b)$

*	1	2	3	6	12
1	1	1	1	1	1
2	1	2	1	2	2
3	1	1	3	3	3
6	1	2	3	6	6
12	1	2	3	6	12

Closure Property : Since all the elements of the table $\in S$, closure property is satisfied.

Associative Property : Since

$$a * (b * c) = a * (b * c) = a * GCD\{b, c\} = GCD\{a, b, c\}$$

$$\text{And } (a * b) * c = GCD\{a, b\} * c = GCD\{a, b, c\}$$

$$\therefore a * (b * c) = (a * b) * c$$

$\therefore *$ is associative.

$\therefore (S, *)$ is a semigroup.

Existence of identity: From the table we observe that $12 \in S$ is the identity

$\therefore (S, *)$ is a monoid.

Commutative property : Since $GCD\{a, b\} = GCD\{b, a\}$ we have $a * b = b * a$. Hence $*$ is commutative.

Therefore A is commutative monoid

(iii) Set $S = \{1, 2, 3, 6, 9, 18\}$ where $a * b = L.C.M. (a, b)$

*	1	2	3	6	9	18
1	1	2	3	6	9	18
2	2	2	6	6	18	18
3	3	6	3	6	9	18
6	6	6	6	6	18	18
9	9	18	9	18	9	18
18	18	18	18	18	18	18

Closure Property : Since all the elements of the table $\in S$, closure property is satisfied.

Associative Property : Since $a * (b * c) = a * LCM\{b, c\} = LCM\{a, b, c\}$

$$\text{And } (a * b) * c = LCM\{a, b\} * c = LCM\{a, b, c\}$$

$$\therefore a * (b * c) = (a * b) * c$$

$\therefore *$ is associative.

Existence of identity : From the table we observe that $1 \in S$ is the identity.

$\therefore (S, *)$ is a monoid.

Commutative property : Since $\text{LCM}\{a, b\} = \text{LCM}\{b, a\}$ we have $a * b = b * a$. Hence $*$ is commutative.

Therefore A is commutative monoid.

(iv) Z , the set of integers where $a * b = a + b - ab$

Closure Property : $a, b \in Z$ then $a + b - ab \in Z \quad \forall a, b$
so $*$ is closure.

Associate Property : Consider $a, b \in Z$

$$\begin{aligned}(a * b) * c &= (a + b - ab) * c \\ &= a + b - ab + c - (a + b - ab)c \\ &= a + b - ab + c - ac - bc + abc \\ &= a + b + c - ab - ac - bc + abc\end{aligned} \quad (1)$$

$$\begin{aligned}a * (b * c) &= a * (b + c - bc) \\ &= a + b + c - bc - a(b + c - bc) \\ &= a + b + c - bc - ab - ac + abc\end{aligned}$$

(2) From 1 & 2

$$(a * b) * c = a * (b * c) \quad \forall a, b, c \in Z$$

$\therefore *$ is associative

$\therefore (Z, &)$ is a semigroup.

Existence of Identity : Let e be the identity element $a * e = a$

$$a + e - a.e = a$$

$$a + e - a.e = a$$

$$e(1-a) = 0$$

$$e = 0 \text{ or } a = 1$$

But $a \neq 1$

$$E = 0$$

$\therefore 0 \in Z$ is the identity element.

$\therefore (Z, *)$ is monoid.

Commutative property : $\forall a, b \in Z$

$$a * b = a + b - ab$$

$$= b + a - ba$$

$$= b * a$$

$\therefore *$ is commutative

$\therefore (Z, *)$ is commutative monoid.

v) $E = \text{set of even integers. } a * b = \frac{ab}{2}$

Closure Property : Since

$\frac{ab}{2}$ is even for a and b even. $\therefore a * b \in E$. Hence
closure property is verified.

Property : Since $a * (b * c) = a * \left(\frac{bc}{2}\right) = \frac{abc}{4} = \frac{ab}{2} * c = (a * b) * c$

$\therefore *$ is associative. $\therefore (E, *)$ is a semigroup.

Existence of identity : $2 \in E$ is the identity because $2 * a = \frac{2a}{2} = a \quad \forall a \in E$

$\therefore (E, *)$ is a monoid.

Commutative property : Since $\frac{ab}{2} = \frac{ba}{2}$, we have $a * b = b * a$ Hence $*$ is commutative.

$\therefore (E, *)$ is commutative monoid.

(vi) $-2 \in A$ is identity

(vii) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \in M$ is the identity

Example 4 : State and prove right or left cancellation property for a group.

Solution : Let $(G, *)$ be a group.

(i) To prove the right cancellation law i.e. $a * b = c * b \Rightarrow a = c$

Let a, b, c $\in G$. Since G is a group, every element has inverse in G.

$\therefore b^{-1} \in G$

Consider $a * b = c * b$

Multiply both sides by b^{-1} from the right.

$\therefore (a * b) * b^{-1} = (c * b) * b^{-1}$

$\therefore a * (b * b^{-1}) = c * (b * b^{-1})$ Associative property

$\therefore e * a = e * c \quad b * b^{-1} = e \in G$

$\therefore a = c$ $e \in G$ is the identity

(ii) To prove the left cancellation law i.e. $a * b = c * b \Rightarrow a = c$

Let $a, b, c \in G$: Since G is a group, every element has inverse in G .

$$\therefore a^{-1} \in G$$

Consider $a * b = a * c$

Multiply both sides by a^{-1} from the left

$$\therefore a^{-1} * (a * b) = a^{-1} * (a * c)$$

$$\therefore (a^{-1} * a) * b = (a^{-1} * a) * c \quad \text{Associative property}$$

$$\therefore e * b = e * c \quad a^{-1} * a = e \in G$$

$$\therefore b = c \quad e \in G \text{ is the identity}$$

Example 5 : Prove the following results for a group G .

(i) The identity element is unique.

(ii) Each a in G has unique inverse a^{-1}

(iii) $(ab)^{-1} = b^{-1}a^{-1}$

Solution : (i) Let G be a group. Let e_1 and e_2 be two identity elements of G .

$$\text{If } e_1 \text{ is identity element then } e_1 e_2 = e_2 e_1 = e_2 \dots\dots\dots(1)$$

$$\text{If } e_2 \text{ is identity element then } e_1 e_2 = e_2 e_1 = e_1 \dots\dots\dots(2)$$

$\therefore$ From (1) and (2) we get $e_1 = e_2$ i.e. identity element is unique.

(ii) Let G be a group. Let b and c be two inverses of $a \in G$.

$$\text{If } b \text{ is an inverse of } a \text{ then } ab = ba = e \dots\dots\dots(1)$$

$$\text{If } c \text{ is an inverse of } a \text{ then } ac = ca = e \dots\dots\dots(2)$$

Where $e \in G$ be the identity element.

$$\therefore \text{ From (1) and (2) we get } ab = ac \text{ and } ba = ca.$$

$$\therefore b=c \text{ by cancellation law : i.e. inverse of } a \in G \text{ is unique.}$$

$$\therefore \text{ inverse of } a \in G \text{ is unique.}$$

(iii) Let G be a group. Let $a, b \in G$.

$$\text{Consider } (ab)(b^{-1}a^{-1})$$

$$= a(bb^{-1})a^{-1} \quad \text{Associative property}$$

$$= (ae)a^{-1} \quad bb^{-1} = e, e \in G \text{ is identity}$$

$$= (ae)a^{-1} \quad \text{Associative property}$$

$$= aa^{-1}e = a$$

Similarly we can prove $(b^{-1}a^{-1})(ab) = e$.

Hence $(ab)^{-1} = b^{-1}a^{-1}$

Example 6 : Let G be a group with identity e . Show that if $a^2 = e$ for all a in G , then every element is its own inverse

Solution : Let G be a group.

Given $a^2 = e$ for all $a \in G$. Multiply by a^{-1}

we get $a^{-1}a^2 = a^{-1}e$

$\therefore a = a^{-1}$

i.e. every element is its own inverse

Example 7 : Show that if every element in a group is its own inverse, then the group must be abelian.

OR

Let G be a group with identity e . Show that if $a^2 = e$ for all a in G , then G is abelian.

Solution : Let G be a group.

$\therefore$ For $a \in G$, $a^{-1} \in G$

$\therefore$ Consider $(ab)^{-1}$

$\therefore (ab)^{-1} = b^{-1}a^{-1}$ reversal law of inverse.

$\therefore ab = ba$ every element is its own inverse

$\therefore G$ is abelian.

Example 8 : Let Z_n denote the set of integers $(0, 1, \dots, n-1)$. Let $\otimes$ be binary operation on Z_n such that $a \otimes b =$ the remainder of ab divided by n .

- i) Construct the table for the operation $\otimes$ for $n=4$.
- ii) Show that $(Z_n, \otimes)$ is a semi-group for any n .
- iii) Is $(Z_n, \otimes)$ a group for any n ? Justify your answer.

Solution : (i) Table for the operation $\otimes$ for $n = 4$.

$\otimes$	0	1	2	3
0	0	0	0	0
1	0	1	2	3
2	0	2	0	2
3	0	3	2	1

- (ii) To show that $(Z_n, \otimes)$ is a semi-group for any n .

Associative property : Since multiplication modulo n is associative, associative property is satisfied.

$\therefore (Z_n, \otimes)$ is a semi-group

(iii) $(Z_n, \otimes)$ is not a group for any n .

Example 9 : Consider the group $G = \{1,2,3,4,5,6\}$ under multiplication modulo 7.

(i) Find the multiplication table of G

(ii) Find $2^{-1}, 3^{-1}, 6^{-1}$.

(iii) Find the order of the subgroups generated by 2 and 3.

(iv) Is G cyclic?

Solution : (i) Multiplication table of G

Binary operation $*$ is multiplication modulo 7.

$*$	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2	4	6	1	3	5
3	3	6	2	5	1	4
4	4	1	5	2	6	3
5	5	3	1	6	4	2
6	6	5	4	3	2	1

From the table we observe that $1 \in G$ is identity.

(ii) To find $2^{-1}, 3^{-1}, 6^{-1}$.

From the table we get $2^{-1} = 4, 3^{-1} = 5, 6^{-1} = 6$

iii) To find the order of the subgroups generated by 2.

Consider $2^0 = 1 = \text{Identity}$, $2^1 = 2$; $2^2 = 4$, $2^3 = 1 = \text{Identity}$

$\langle 2 \rangle = \{2^1, 2^2, 2^3\}$

$\therefore$ Order of the subgroup generated by 2 = 3

To find the order of the subgroups generated by 3.

Consider $3^0 = 1 = \text{identity}$, $3^1 = 3$, $3^2 = 2$, $3^3 = 6$, $3^4 = 4$, $3^5 = 5$, $3^6 = 1 = \text{Identity}$

$\langle 3 \rangle = \{3^1, 3^2, 3^3, 3^4, 3^5, 3^6\}$

$\therefore$ Order of the subgroup generated by 3 = 6

(iv) G is cyclic because $G = \langle 3 \rangle$.

Example 10 : Let G be an abelian group with identity e and let $H = \{x/x^2 = e\}$. Show that H is a subgroup of G .

Solution : Let $x, y \in H$ $\therefore x^2 = e$ and $y^2 = e \quad \therefore x^{-1} = x$ and $y^{-1} = y$

Since G is abelian we have $xy = yx \therefore xy^{-1} = yx$

$$\begin{aligned} \text{Now } (xy^{-1})^2 &= (xy^{-1})(xy^{-1}) = (xy^{-1})(y^{-1}x) \\ &= (xy^{-1})(yx) = x(y^{-1}y)x \\ &= x(e)x \\ &= x^2 = e \end{aligned}$$

$$\Rightarrow xy^{-1} \in H$$

$\therefore H$ is a subgroup.

Example 16 : Let G be a group and let $H = \{x/x \in G \text{ and } xy = yx \text{ for all } y \in G\}$. Prove that H is a subgroup of G .

Solution : Let $x, z \in H \therefore xy = yx$ for every $y \in G \therefore x = yxy^{-1}$.

Similarly $zy = yz$ for every $y \in G \therefore z = yzy^{-1}$.

$$\begin{aligned} \text{Now consider } xz^{-1} &= (yxy^{-1})(yzy^{-1})^{-1} \\ &= yxy^{-1}yz^{-1}y^{-1} = yxz^{-1}y^{-1} \end{aligned}$$

$$\Rightarrow (x.z^{-1})y = y(xz^{-1}) \in H.$$

$$\Rightarrow xz^{-1} \in H$$

$\therefore H$ is a subgroup

Example 17 : Find all subgroups of $(Z, \oplus)$ where $\oplus$ is the operation addition modulo 5. Justify your answer.

Solution:

$\oplus$	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

Example 18 : Let G be a group of integers under the operation of addition. Which of the following subsets of G are subgroups of G ?

(a) the set of all even integers,

(b) the set of all odd integers. Justify your answer.

Solution:

- a) Let H = set of all even integers.
We know, additive inverse of an even number is even and sum of two even integers is also even. Thus for $a, b \in H$ we have $ab^{-1} \in H$.
Hence H is a subgroup of G .
- b) Let K = set of all odd integers.
We know, additive inverse of an odd number is odd and sum of two odd integers is even.
Thus for $a, b \in K$ we have $ab^{-1} \notin K$.
Hence K is not a subgroup of G .

Example 19 : Let $(G, *)$ be a group and H be a non-empty subset of G . Show that $(H, *)$ is a subgroup if for any a and b in H , ab^{-1} is also in H .

Solution :

- (i) Let $a, a \in H \therefore a a^{-1} \in H$ i.e. $e \in H$
 $\therefore$ The identity element $\in H$.
- (ii) Let $e, a \in H \therefore ea^{-1} \in H$ i.e. $a^{-1} \in H$
 $\therefore$ Every element has inverse $\in H$.
- (iii) Let $a, b \in H \therefore b^{-1} \in H \therefore a(b^{-1})^{-1} \in H$ i.e. $ab \in H$.
 $\therefore$ Closure property is satisfied.
- (iv) Every element in H is also in G . And G is a group. So associative property is satisfied by the elements of H . Hence associative property is satisfied by the elements of H .
Hence H is a group. But H is a subset of $G \therefore H$ is a subgroup of G .

Example 20 : Let H and K be subgroups of a group G . Prove that $H \cap K$ is a subgroup of G .

Solution : If H is a subgroups of a group G , then for any $a, b \in H$, $ab^{-1} \in H$.

Similarly, if K is a subgroups of a group G , then for any $a, b \in K$, $ab^{-1} \in K$.

Now if $a, b \in H \cap K$, $a, b \in H$ and $a, b \in K \therefore ab^{-1} \in H$ and $ab^{-1} \in K$.
Hence $ab^{-1} \in H \cap K$.

$\therefore H \cap K$ is a subgroup of G .

HOMOMORPHISM, ISOMORPHISM AND AUTOMORPHISM OF SEMIGROUPS

Homomorphism : Let $(S, *)$ and $(T, *')$ be two semigroups. An everywhere defined function

$f: S \rightarrow T$ is called a homomorphism from $(S, *)$ to $(T, *')$ if

$$f(a * b) = f(a) *' f(b) \quad \forall a, b \in S$$

Isomorphism : Let $(S, *)$ and $(T, *')$ be two semigroups. A function

$f: S \rightarrow T$ is called a isomorphism from $(S, *)$ to $(T, *')$ if

(i) it is one-to-one correspondence from S to T (ii) $f(a * b) = f(a) *' f(b) \quad \forall a, b \in S$

$(S, *)$ and $(T, *')$ are isomorphic' is denoted by $S \cong T$.

Automorphism : An isomorphism from a semigroup to itself is called an automorphism of the semigroup. An isomorphism $f: S \rightarrow S$ is called automorphism.

HOMOMORPHISM, ISOMORPHISM AND AUTOMORPHISM OF MONOIDS :

Homomorphism : Let $(M, *)$ and $(M', *')$ be two monoids. An everywhere defined function $f: M \rightarrow M'$ is called a homomorphism from $(M, *)$ to $(M', *')$ if $f(a * b) = f(a) *' f(b) \quad \forall a, b \in M$

Isomorphism : Let $(M, *)$ and $(M', *')$ be two monoids. A function

$f: M \rightarrow M'$ is called a isomorphism from $(M, *)$ to $(M', *')$ if

(i) it is one-to-one correspondence from M to M' (ii) f is onto.

(iii) $f(a * b) = f(a) *' f(b) \quad \forall a, b \in M$

$(M, *)$ and $(M', *')$ are isomorphic is denoted by $M \cong M'$.

Automorphism : An isomorphism from a monoid to itself is called an automorphism of the monoid. An isomorphism $f: M \rightarrow M$ is called automorphism of monoid.

HOMOMORPHISM, ISOMORPHISM AND AUTOMORPHISM OF GROUPS :

Homomorphism : Let $(G, *)$ and $(G', *')$ be two groups. An everywhere defined function $f: G \rightarrow G'$ is called a homomorphism from $(G, *)$ to $(G', *')$ if

$$f(a * b) = f(a) *' f(b) \quad \forall a, b \in G$$

Isomorphism : Let $(G, *)$ and $(G', *')$ be two groups. A function $f: G \rightarrow G'$ is called a isomorphism from $(G, *)$ to $(G', *')$ if

(i) it is one-to-one correspondence from G to G' (ii) f is onto.

$$(iii) \quad f(a * b) = f(a) *' f(b) \quad \forall a, b \in G$$

' $(G, *)$ and $(G', *')$ are isomorphic' is denoted by $G \cong G'$.

Automorahism: An isomorphism from a group to itself is called an automorphism of the group. An isomorphism $f: G \rightarrow G$ is called Automorphism

Theorem : Let $(S, *)$ and $(T, *')$ be monoids with identity e and e' , respectively. Let $f: S \rightarrow T$ be an isomorphism. Then $f(e) = e'$.

Proof : Let b be any element of T . Since f is on to, there is an element a in S such that $f(a) = b$

Then $a = a * e$

$$b = f(a) = f(a * e) = f(a) * f(e) = b *' f(e) \quad (f \text{ is isomorphism})$$

Similarly, since $a = e * a$,

$$b = f(a) = f(e * a) = f(e) *' f(a) = f(e) *' b$$

Thus for any $b \in T$,

$$b = b *' f(e) = f(e) *' b$$

which means that $f(e)$ is an identity for T .

Thus since the identity is unique, it follows that $f(e) = e'$

Theorem : Let f be a homomorphism from a semigroup $(S, *)$ to a semigroup $(T, *')$. If S' is a subsemigroup of $(S, *)$, then $F(S') = \{t \in T \mid t = f(s) \text{ for some } s \in S'\}$, The image of S' under f , is subsemigroup of $(T, *')$.

Proof : If t_1 , and t_2 are any elements of $F(S')$, then there exist s_1 and s_2 in S' with $t_1 = f(s_1)$ and $t_2 = f(s_2)$. Therefore,

$$t_1 *' t_2 = f(s_1) *' f(s_2) = f(s_1 * s_2) = f(s_2 * s_1) = f(s_2) *' f(s_1) = t_2 *' t_1$$

Hence $(T, *')$ is also commutative.

Example 1 : Let G be a group. Show that the function $f: G \rightarrow G$ defined by $f(a) = a^2$ is a homomorphism iff G is abelian.

Solution :

Step-1: Assume G is abelian. Prove that $f : G \rightarrow G$ defined by $f(a) = a^2$ is a homomorphism.

Let $a, b \in G$. $\therefore f(a) = a^2$, $f(b) = b^2$ and $f(ab) = (ab)^2$ by definition of f .

$$\begin{aligned}
 \therefore f(ab) &= (ab)^2 \\
 &= (ab)(ab). \\
 &= a(ba)b && \text{associativity} \\
 &= a(ab)b && G \text{ is abelian} \\
 &= (aa)(bb) && \text{associativity} \\
 &= a^2 b^2 \\
 &= f(a)f(b) && \text{definition of } f
 \end{aligned}$$

$\therefore f$ is a homomorphism.

Step 2 :

$$\forall y = a^2 \in G \exists a \in G \text{ s.t.}$$

$$f(a) = y = a^2$$

$\therefore f$ is onto.

Step-3: Assume, $f : G \rightarrow G$ defined by $f(a) = a^2$ is a homomorphism. Prove that G is abelian.

Let $a, b \in G$. $\therefore f(a) = a^2$, $f(b) = b^2$ and $f(ab) = (ab)^2$ by definition of f .

$$\begin{aligned}
 \therefore f(ab) &= f(a)f(b) && f \text{ is homomorphism} \\
 \therefore (ab)^2 &= a^2 b^2 && \text{definition of } f \\
 \therefore (ab)(ab) &= (aa)(bb) \\
 \therefore a(ba)b &= a(ab)b && \text{associativity} \\
 \therefore ba &= ab && \text{left and right cancellation laws} \\
 \therefore G &\text{ is abelian.}
 \end{aligned}$$

Example 3 : Let G be a group and let a be a fixed element of G . Show that the function $f_a : G \rightarrow G$ defined by $f_a(x) = axa^{-1}$ for $x \in G$ is an isomorphism.

Solution :

Step-1: Show that f is 1-1.

$$f_a(x) = axa^{-1}$$

Consider $f_a(x) = f_a(y)$ for $x, y \in G$

$$\therefore axa^{-1} = aya^{-1} \quad \text{definition of } f$$

$$\therefore x = y \quad \text{left and right cancellation laws}$$

$$\therefore f \text{ is 1-1}$$

Step 2 :

$$\forall y = axa^{-1} \in G \exists x \in G \text{ s.t.}$$

$$f_a(x) = axa^{-1}$$

$\therefore f$ is onto.

Step-3: Show that f is homomorphism.

For $x, y \in G$

$$f(x) = a * x * a^{-1}, \quad f(y) = a * y * a^{-1} \quad \text{and} \quad f(x * y) = a * (x * y) * a^{-1}$$

Consider $f(x * y) = a * (x * y) * a^{-1}$ for $x, y \in G$

$$\therefore f(x * y) = a * (x * e * y) * a^{-1} \quad e \in G \text{ is identity}$$

$$= a * (x * a^{-1} * a * y) * a^{-1} \quad a^{-1} * a = e$$

$$= (a * x * a^{-1}) * (a * y * a^{-1}) \text{ associativity}$$

$$\therefore *f(x * y) = f(x) * f(y)$$

$\therefore f$ is homomorphism.

Since f is 1-1 and homomorphism, it is isomorphism.

Example 2 : Let G be a group. Show that the function $f : G \rightarrow G$ defined by $f(a) = a^{-1}$ is an isomorphism if and only if G is abelian.

Solution :

Step-1: Assume G is abelian. Prove that $f : G \rightarrow G$ defined by $f(a) = a^{-1}$ is an isomorphism.

i) Let $f(a)=f(b)$

$$\therefore a^{-1} = b^{-1} \quad \therefore a = b \quad \therefore f \text{ is 1-1.}$$

ii) $\forall a \in G \Rightarrow a^{-1} \in G$

$$\therefore x^{-1} \in G$$

$$\Rightarrow f(x) = x^{-1}$$

$\therefore f$ is onto.

iii) Let $a, b \in G$. $\therefore f(a) = a^{-1}$, $f(b) = b^{-1}$ and $f(ab) = (ab)^{-1}$ by definition of f .

$$\begin{aligned} \therefore f(ab) &= (ab)^{-1} \\ &= b^{-1}a^{-1} && \text{reversal law of inverse} \\ &= a^{-1}b^{-1} && G \text{ is abelian} \\ &= f(a)f(b) && \text{definition of } f. \end{aligned}$$

$\therefore f$ is a homomorphism.

Since f is 1-1 and homomorphism, it is isomorphism.

Step – 2: Assume $f : G \rightarrow G$ defined by $f(a) = a^{-1}$ is an isomorphism. Prove that G is abelian.

Let $a, b \in G$ $\therefore f(a) = a^{-1}$, $f(b) = b^{-1}$ and $f(ab) = (ab)^{-1}$ by definition of f

$$\therefore f(ab) = f(a)f(b) \quad f \text{ is homomorphism}$$

$$\therefore (ab)^{-1} = a^{-1}b^{-1} \quad \text{definition of } f$$

$$\therefore b^{-1}a^{-1} = a^{-1}b^{-1} \quad \text{reversal law of inverse}$$

G is abelian.

Example 3 : Define $(\mathbb{Z}, +) \rightarrow (5\mathbb{Z}, +)$ as $f(x) = 5x$, where $5\mathbb{Z} = \{5n : n \in \mathbb{Z}\}$. Verify that f is an isomorphism.

Solution:

Step -1 Show that f is 1-1.

Consider $f(x) = f(y)$ for $x, y \in G$

$$\therefore 5x = 5y \quad \text{definition of } f$$

$$\therefore x = y \quad \therefore f \text{ is 1-1}$$

Step 2 :

$$\forall 5x \in G, \exists x \in G$$

$$\text{s.t. } f(x) = 5x$$

$\therefore f$ is onto.

Step-3: Show that f is homomorphism.

For $x * y \in G$

$$f(x) = 5x, f(y) = 5y \text{ and } f(x + y) = 5(x+y)$$

$$\text{Consider } f(x+y) = 5(x+y) \quad \text{for } x, y \in G \\ = 5x + 5y$$

$$\therefore f(x+y) = f(x) + f(y)$$

$\therefore f$ is homomorphism.

Since f is 1-1 and homomorphism, it is isomorphism.

Example 4 : Let G be a group of real numbers under addition, and let G' be the group of positive numbers under multiplication. Let $f : G \rightarrow G'$ be defined by $f(x) = e^x$. Show that f is an isomorphism from G to G'

OR

Show that the group $G = (R, +)$ is isomorphic to $G' = (R^+, \cdot)$ where R is the set of real numbers and R^+ is a set of positive real numbers.

Solution :

Step 1: Show that f is 1-1.

$$\text{Consider } f(x) = f(y) \quad \text{for } x, y \in G$$

$$\therefore e^x = e^y \quad \text{definition of } f$$

$$\therefore x = y \quad \therefore f \text{ is 1-1.}$$

Step 2 : If $x \in G^1$, then $\log x \in G$ and $f(\log x) = e^{\log x} = x$ so f is onto.

Step-3: Show that f is homomorphism.

For $x, y \in G$

$$f(x) = e^x, f(y) = e^y \text{ and } f(x+y) = e^{(x+y)}$$

$$\begin{aligned} \text{Consider } f(x+y) &= e^{(x+y)} \text{ for } x, y \in G \\ &= e^x \times e^y \end{aligned}$$

$$\therefore f(x+y) = f(x) \times f(y) \quad f \text{ is homomorphism.}$$

Since f is 1-1 and homomorphism, it is isomorphism.

Example 5 : Let $G = \{e, a, a^2, a^3, a^4, a^5\}$ be a group under the operation of $a^i a^j = a^r$, where $i+j \equiv r \pmod{6}$. Prove that G and Z_6 are isomorphic

Solution :

Step - I: Show that f is 1-1.

Let $x = a^i$, and $y = a^j$.

Consider $f(x) = f(y)$ for $x, y \in G$

$$\therefore f(a^i) = f(a^j) \quad \text{definition of } f$$

$$\therefore a^i = a^j$$

$$\therefore x = y \quad f \text{ is 1-1.}$$

Step-2: Show that f is homomorphism.

Let $x = a^i$ and $y = a^j$, $x, y \in G$

$$f(a^i) = i, f(a^j) = j \text{ and } f(x+y) = f(a^i a^j)$$

$$\text{Consider } f(x+y) = f(a^i a^j) = f(a^r) \quad \text{where } i+j \equiv r \pmod{6}$$

$$= R$$

$$= i+j$$

$$= f(a^i) + f(a^j)$$

$$\therefore f(x+y) = f(x) + f(y) \quad \therefore f \text{ is homomorphism.}$$

Since f is 1-1 and homomorphism, it is isomorphism.

Example 6 : Let T be set of even integers. Show that the semigroups $(Z, +)$ and $(T, +)$ are isomorphic.

Solution : We show that f is one to one onto .

Define $f: (Z, +) \rightarrow (T, +)$ as $f(x) = 2x$

1) Show that f is 1-1

Consider $f(x) = f(y)$

$$\therefore 2x = 2y$$

$$\therefore x = y \quad \therefore f \text{ is 1-1.}$$

2) Show that f is onto

$$y = 2x \quad \therefore x = y/2 \text{ when } y \text{ is even.}$$

$$\therefore \text{for every } y \in T \text{ there exists } x \in Z.$$

$$\therefore f \text{ is onto.}$$

$$\therefore f \text{ is isomorphic.}$$

- 3) F is homomorphism

$$F(x + y) = 2(x + y)$$

$$= 2x + 2y$$

$$= f(x) + f(y)$$

$$\therefore f \text{ is homomorphism.}$$

Example 7 : For the set $A = \{a, b, c\}$ give all the permutations of A. Show that the set of all permutations of A is a group under the composition operation.

Solution : $A = \{a, b, c\}$. S_3 = Set of all permutations of A.

$$f_0 = \begin{pmatrix} a & b & c \\ a & b & c \end{pmatrix}, \quad f_1 = \begin{pmatrix} a & b & c \\ a & c & b \end{pmatrix}, \quad f_2 = \begin{pmatrix} a & b & c \\ c & b & a \end{pmatrix}$$

$$f_3 = \begin{pmatrix} a & b & c \\ b & a & c \end{pmatrix}, \quad f_4 = \begin{pmatrix} a & b & c \\ b & c & a \end{pmatrix}, \quad f_5 = \begin{pmatrix} a & b & c \\ c & a & b \end{pmatrix}$$

Let us prepare the composition table.

0	f_0	f_1	f_2	f_3	f_4	f_5
f_0	f_0	f_1	f_2	f_3	f_4	f_5
f_1	f_1	f_0	f_4	f_5	f_2	f_3
f_2	f_2	f_3	f_0	f_4	f_3	f_1
f_3	f_3	f_4	f_5	f_0	f_1	f_2
f_4	f_4	f_3	f_1	f_2	f_5	f_0
f_5	f_5	f_2	f_3	f_1	f_0	f_4

- Closure Property:** Since all the elements in the composition table $\in S_3$, closure property is satisfied.
- Associative Property:** Since composition of permutations is associative, associative property is satisfied.
- Existence of Identity:** From the table we find that f_0 is the identity
- Existence of Inverse:** From the composition table it is clear that $f_0^{-1} = f_0$, $f_1^{-1} = f_1$, $f_2^{-1} = f_2$, $f_3^{-1} = f_3$, $f_4^{-1} = f_5$, $f_5^{-1} = f_4$
 $\therefore$ Every element has inverse in S_3 . Hence S_3 is a group.

COSET AND NORMAL SUBGROUP:

Left Coset : Let $(H, *)$ be a subgroup of $(G, *)$. For any $a \in G$, the set of aH defined by $aH = \{a * h / h \in H\}$ is called the **left coset** of H in G

determined by the element $a \in G$. The element a is called the representative element of the left coset aH .

Right Coset : Let $(H, *)$ be a subgroup of $(G, *)$. For any $a \in G$, the set of Ha defined by

$$Ha = [h * a \mid h \in H]$$

is called the **right coset** of H in G determined by the element $a \in G$. The element a is called the representative element of the right coset Ha .

Theorem : Let $(H, *)$ be a subgroup of $(G, *)$. The set of left cosets of H in G form a partition of G . Every element of G belongs to one and only one left coset of H in G .

Lagrange' Theorem: The order of a subgroup of a finite group divides the order of the group.

Corollary : If $(G, *)$ is a finite group of order n , then for any $a \in G$, we must have $a^n = e$, where e is the identity of the group.

Normal Subgroup : A subgroup $(H, *)$ of $(G, *)$ is called a normal subgroup if for any $a \in G$, $aH = Ha$.

Example 8 : Determine all the proper subgroups of symmetric group (S_3, o) . Which of these subgroups are normal?

Solution : $S = \{1, 2, 3\}$. $S_3 =$ Set of all permutations of S .

$S_3 = \{f_0, f_1, f_2, f_3, f_4, f_5\}$ where

$$\begin{aligned} f_0 &= \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}, & f_1 &= \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}, & f_3 &= \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \\ f_2 &= \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}, & f_4 &= \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}, & f_5 &= \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \end{aligned}$$

Let us prepare the composition table.

0	f_0	f_1	f_2	f_3	f_4	f_5
f_0	f_0	f_1	f_2	f_3	f_4	f_5
f_1	f_1	f_0	f_4	f_5	f_2	f_3
f_2	f_2	f_3	f_0	f_4	f_3	f_1
f_3	f_3	f_4	f_5	f_0	f_1	f_2
f_4	f_4	f_3	f_1	f_2	f_5	f_0
f_5	f_5	f_2	f_3	f_1	f_0	f_4

From the table it is clear that $\{f_0, f_1\}$, $\{f_0, f_2\}$, $\{f_0, f_3\}$ and $\{f_0, f_4, f_5\}$ are subgroups of $(S_3, 0)$: The left cosets of $\{f_0, f_1\}$ are $\{f_0, f_1\}$, $\{f_2, f_5\}$, $\{f_3, f_4\}$. While the right cosets of $\{f_0, f_1\}$ are $\{f_0, f_1\}$, $\{f_2, f_4\}$, $\{f_3, f_5\}$. Hence $\{f_0, f_1\}$ is not a normal subgroup.

Similarly we can show that $\{f_0, f_2\}$ and $\{f_0, f_3\}$ are not normal subgroups.

On the other hand, the left and right cosets of $\{f_0, f_4, f_5\}$ are $\{f_0, f_4, f_5\}$ and $\{f_1, f_2, f_3\}$.

Hence $\{f_0, f_4, f_5\}$ is a normal subgroup.

Example 9: Let $S = \{1, 2, 3\}$. Let $G = S_3$ be the group of all permutations of elements of S , under the operation of composition of permutations.

Let H be the subgroup formed by the two permutations $\begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}$ and

$\begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$. Find the left coset of H in G . Is H a normal subgroup? Explain your notion of composition clearly.

Solution : Let

$$f_0 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}, \quad f_1 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}, \quad f_3 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$$

$$f_2 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}, \quad f_4 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}, \quad f_5 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$$

$$\therefore H = \{f_0, f_2\}$$

Left Cosets of H in G:

$$\begin{aligned}
f_0H &= \{f_0f_0, f_0f_2\} = \{f_0, f_2\} & f_1H &= \{f_1f_0, f_1f_2\} = \{f_1, f_4\} \\
f_2H &= \{f_2f_0, f_2f_2\} = \{f_2, f_0\} & f_3H &= \{f_3f_0, f_3f_2\} = \{f_3, f_5\} \\
f_4H &= \{f_4f_0, f_4f_2\} = \{f_4, f_1\} & f_5H &= \{f_5f_0, f_5f_2\} = \{f_5, f_3\}
\end{aligned}$$

Right Cosets of H in G

$$Hf_0 = \{f_0f_0, f_2f_0\} = \{f_0, f_2\} \quad Hf_1 = \{f_0f_1, f_2f_1\} = \{f_1, f_3\}$$

Since $f_1H \neq Hf_1$, H is not a normal subgroup of G.

Example 10 : Define a normal sub-group. Let S_3 = Group of all permutations of 3 elements (say 1, 2, 3). For the following subgroups of S_3 , find all the left cosets. Subgroup of A = $\{1, (1,2)\}$

Where I = identity permutation, (1, 2) is a transposition. Is A a normal subgroup. State a normal subgroup of the above group if it exists.

Solution : $H = \{f_0, f_3\}$

The left cosets of H in G are as follow.

$$\begin{aligned}
f_0H &= \{f_0, f_3\} & f_1H &= \{f_1, f_5\} & f_2H &= \{f_2, f_4\} \\
f_3H &= \{f_3, f_0\} & f_4H &= \{f_4, f_2\} & f_5H &= \{f_5, f_1\}
\end{aligned}$$

Consider a right coset $Hf_1 = \{f_1, f_4\}$

Since $f_1H \neq Hf_1$, H is not a normal subgroup of G.

RING: An algebraic structure $(R, +, \cdot)$ is said to be a Ring if it satisfies :

- $(R, +)$ is a commutative Group.
- $(R, \cdot)$ is a semigroup and
- $(R, +, \cdot)$ satisfies the distributive property.

FIELD: An algebraic structure $(F, +, \cdot)$ is said to be a Field if it satisfies :

- $(F, +)$ is a commutative Group.
- $(F, \cdot)$ is a commutative group and
- $(F, +, \cdot)$ satisfies the distributive property.

Zero Divisor: A commutative ring is said to have a zero divisor if the product of two non-zero element is zero. For example, the product of two non- zero matrices may zero.

INTEGRAL DOMAIN: A commutative without a zero divisor is called an integral domain.

THEOREM: Every finite integral domain is a field.

THEOREM: Every field is an integral domain.

Unit IV

LATTICE THEORY, BOOLEAN ALGEBRA AND CODING THEORY

OBJECTIVES:

After going through this unit, you will be able to :

- Define basic terminology associated with lattice theory.
- Boolean lattices and Boolean algebras
- Coding theory

LATTICES

BASIC TERMINOLOGY

Definition:

A poset is a **lattice** if every pair of elements has a lub (join) and a glb (meet).

Least upper bound (lub)

Let $(A, \leq)$ be a poset and B be a subset of A .

1. An element $a \in A$ is an upper bound for B iff for every element $a' \in B$, $a' \leq a$.
2. An element $a \in A$ is a least upper bound (lub) for B iff a is an upper bound for B and for every upper bound a' for B , $a \leq a'$.

Greatest lower bound (glb)

Let $(A, \leq)$ be a poset and B be a subset of A .

1. An element $a \in A$ is a lower bound for B iff for every element $a' \in B$, $a \leq a'$.
2. An element $a \in A$ is a greatest lower bound (glb) for B iff a is a lower bound for B and for every lower bound a' for B , $a' \leq a$.

Theorem:

Let $(L, \leq)$ be a lattice, For any $a, b, c \in L$,

- (i) $a * a = a$ (i') $a + a = a$ (idempotent)
- (ii) $a * b = b * a$ (ii') $a + b = b + a$ (Commutative)